Factoring with special products

1 Write an expression that makes each of the following statements true.

a $p^2 - 64 = (p - 8)(\square)$

b $m^2 - 36 = (m - 6)\,(\square)$

2 Find the other factor of $x^2 - 49$ if one of its factors is $x + 7$.

3 Factor $x^4 - 100$ as a product of two binomials.

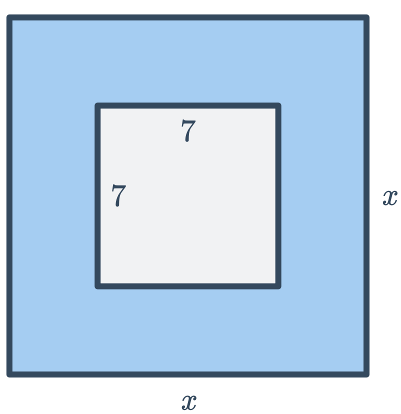
4 Consider the expression $81x^2 - 225$
Write the factored form of an equivalent expression, where a and b are integers.

5 Factor the following expressions:

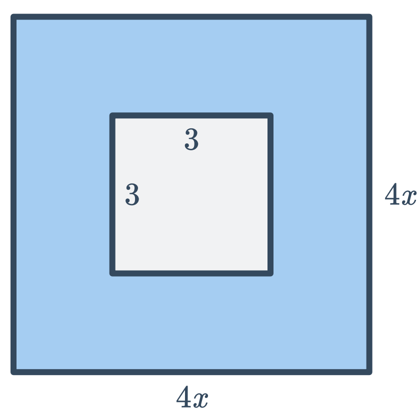
a $x^2 - 25$	b $n^2 - 36$	c $v^2 - 49$	d $x^2 - \dfrac{4}{49}$
e $81 - x^2$	f $4 - u^2$	g $4 - 49y^2$	h $64x^2 - 121$
i $121m^2 - 64$	j $81x^2 - 100y^2$	k $121x^2 - 49y^2$	l $16x^2 - 25$
m $6x^2 - 294$	n $3t^2 - 12$	o $5x^2 - 320$	p $4n^2 - 16$

6 For the following diagrams, find the area of the shaded region. Express your answer in factored form.

a



b



7 Evaluate $9x^2 - 4$ when $x = 4$ by first factoring the expression.

8 Find the value of $98^2 - 96^2$ using the difference of 2 squares identity.

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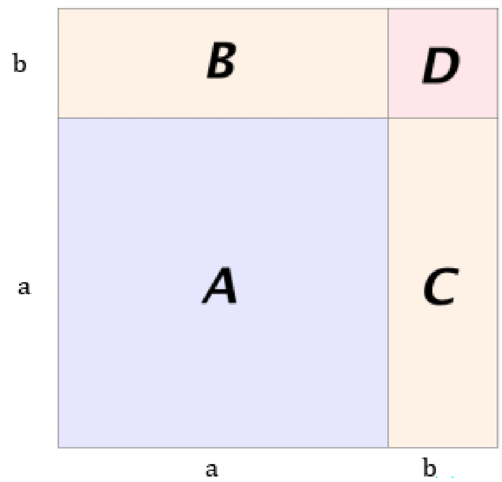
10 Consider the square shown in the diagram below, which has been divided into four rectangular sections.



a Find the area of the whole square in simplest factored form.

b Find the area of:

i Rectangle A ii Rectangle B

iii Rectangle C iv Rectangle D

c Using the areas of each rectangle, write a different expression for the area of the whole square.

d Name the identity that corresponds to your answers in parts (a) and (c).

11 Factor the following:

a $b^2 + 14b + 49$

b $y^2 - 10y + 25$

c $x^2 + 10x + 25$

d $x^2 + 14x + 49$

e $x^2 - 12x + 36$

f $x^2 - 16x + 64$

g $81x^2 - 72x + 16$

h $81x^2 + 36x + 4$

i $49x^2 - 28x + 4$

j $81 + 18x + x^2$

k $36 - 12x + x^2$

l $25 + 60r + 36r^2$

m $7d^2 + 112d + 448$

n $4x^2 + 40x + 100$

o $3x^2 - 24xy + 48y^2$

p $-6r^2 - 96r - 384.$

q $-3x^2 + 12x - 12$

r $b^2 - 7b + \frac{49}{4}$

s $x^2 + 3x + \frac{9}{4}$

t $x^2 - 3x + \frac{9}{4}$

12 For each of the following scenarios:

i Factor the expression completely.

ii Find an expression for the length of the side.

a A square has an area of $x^2 + 12x + 36.$

b Three identical squares have a total area of $12x^2 + 12x + 3.$

c A cube has a surface area of $6x^2 + 36x + 54.$

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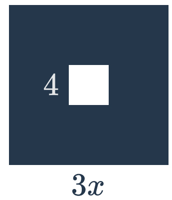
13 Find the value of k that will make $16x^2 - 24$[illegible] a perfect square trinomial.

Additional questions

14 A shape is formed from a square with side lengths of $3x$ has a smaller square, with side lengths of 4, cut out from the center.



a Write a simplified expression for the area of this shape.

b A rectangle is created to have the same area found in part (a). Determine the dimensions for the rectangle.

15 Using the digits 1 to 9 with no repeats, fill in the blanks to create a perfect square trinomial:
$\square x^2 + \square\square x + \square\square$

16 Rewrite the expression $\left(a^2 - b^2\right)\left(c^2 - d^2\right)$ as a difference of two squares.

17 Sheldon and Gabriella each factored $16x^2 + 48x + 36$.
Whose work is incorrect? Identify and explain the error.

Sheldon:

$$\begin{aligned} 16x^2 + 48x + 36 &= 16x^2 + 24x + 24x + 36 \\ &= 8x(2x+3) + 12(2x+3) \\ &= (8x+12)(2x+3) \\ &= 4(2x+3)(2x+3) \end{aligned}$$

Gabriella:

$$\begin{aligned} 16x^2 + 48x + 36 &= (4x+6)(4x+6) \\ &= 2(2x+3)(2x+3) \end{aligned}$$

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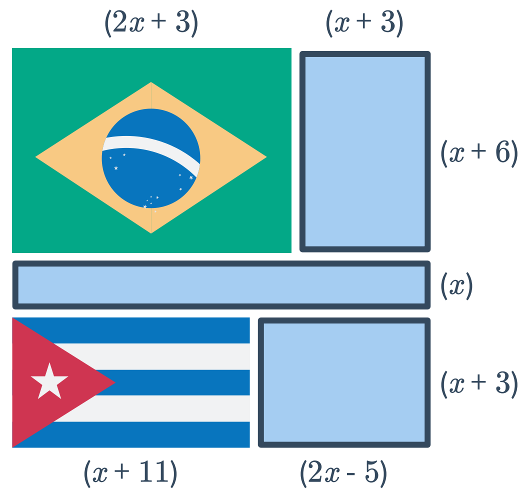
18 Maribel is crafting a large blanket to give to her parents for their anniversary. Her plan is to combine a patchwork of family photographs with the flags from her parents' home countries. She hasn't determined the exact size of the blanket yet, and instead has expressions for the possible dimensions of each piece of fabric.



a Write an expression for the total area of the blanket. Show at least three possible factorizations for the area.

b Find a value for x that creates a blanket with reasonable dimensions. Be sure to include units.